



Dominant Psychosocial Response & Distress Classification in Oncology Messages

Updated proposal - a novel, synthetic, label-leakage-controlled NLP benchmark

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MOTIVATING USE CASE

Reading the emotional state behind oncology messages

- Background: a cancer diagnosis carries a heavy psychosocial burden; supportive-care teams want to triage emotional state from patient-generated text.
- Why it matters: distress drives care needs. Two signals matter - which response dominates, and how intense the distress is.
- Why it is challenging: no public labeled dataset for this framing, and emotional labels are subjective and expensive to annotate.
- How it is solved today: manual screening (e.g. distress thermometers) and generic sentiment tools that miss fine-grained, mixed affect.

Two questions per message

1) Dominant psychosocial response

anxiety, sadness, anger, hope,
guilt, denial, acceptance

2) Distress intensity (ordinal)

low < medium < high

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PROJECT TASK

Two labels per message

- Formal task: $f(\text{text}) \rightarrow (\text{response in 7 classes, distress in \{low, medium, high\}})$.
- Input: one English oncology-related message. Output: the dominant response and the ordinal distress level (predicted by separate models).
- Novelty: unlike broad sentiment work, we jointly study fine-grained response type and ordinal distress on an attribute-controlled synthetic corpus.

Input -> Output (example)

“Scan is on Thursday. I keep rereading the same line and nothing goes in, and I have checked my phone for the clinic number ten times tonight.” -> anxiety / medium

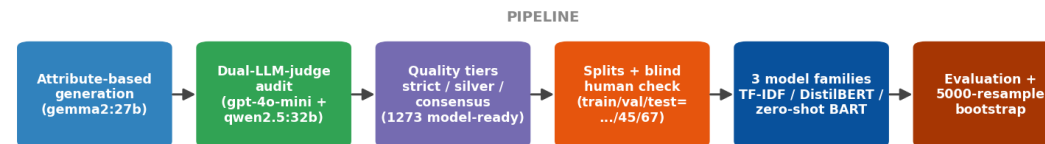
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MODELS & METHODS

Pipeline and three model families

Dominant Psychosocial Response & Distress Classification in Oncology Messages

A fully synthetic, label-leakage-controlled NLP benchmark: 7 psychosocial responses + 3 distress levels



KEY RESULTS (test macro-F1, 67 items)

RESPONSE (7-class)	
TF-IDF-B (best overall)	0.856
DistilBERT-C (best transformer)	0.834
Zero-shot BART	0.732
human-vs-label kappa	0.84
lexical & transformer tied on 67 items	

DISTRESS (3-level)	
DistilBERT-B (best)	0.864
TF-IDF nominal-B	0.805
Zero-shot BART	0.468
human-vs-label wKappa	0.86
both supervised >> zero-shot	

Finding: synthetic supervision helps both tasks, but its value depends on task and model; more consensus data is not uniformly better.

Generator: gemma2:27b (Ollama). Judges: gpt-4o-mini + qwen2.5:32b. Models compared: TF-IDF + Logistic Regression, fine-tuned DistilBERT, zero-shot BART-MNLI.

Fine-tuning: DistilBERT with class-weighted cross-entropy, separate models for response and distress.

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DATA SPECIFICATION & GENERATION

Attribute-based synthetic generation

- Requirements: diverse, realistic, and leakage-free text - the message must never explicitly name the target response label.
- Attribute space: role, cancer stage, cancer type, tone, channel, age group, length, noise - sampled to force coverage.
- Generate: condition the LLM on attributes + an intended label; reject malformed or constraint-violating outputs; retain confident label drift only when both judges agree on a consensus label.
- Label & audit: two blind LLM judges predict the labels; agreement assigns each item to a quality tier.

Tier	Records
strict	471
silver	75
consensus	727
model-ready	1273

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METRICS & KPIs

How we measure success

Primary

macro-F1
(equal weight to all classes)

Response

micro / macro precision,
recall, F1 + minority macro-F1

Distress (ordinal)

weighted kappa, MAE,
severe (low<->high) rate, per-level

Significance

5000-resample paired
bootstrap, 95% CIs

Human check

Cohen / weighted kappa vs.
a blind annotator on the test set

Guardrail

test set never used for tuning
(validation only)